



B3824

Enrollment No: _____

EVEN SEMESTER MAKEUP/SUPPLEMENTARY/SUMMER TERM EXAMINATION, JULY: - 2025**Course Code : CSET301****Course Name : Artificial Intelligence and Machine Learning****Program Name: B.Tech. CSE****Semester: IV****Time: 02:30 Hours****Max Marks: 40****General Instruction: -Do not write anything on the question paper except enrollment number.****Note: - Attempt the questions as per instruction given in each section.****Use of Scientific Calculators are allowed. Log table/Statistical table shall be provided as per requirement.****Section A:** All questions in this section are compulsory. Attempt either A or B part of each question. Each question carries 03 marks. **[5QX3=15 Marks]****Q1.****1A.** Explain RNNs. In what scenarios are RNNs particularly effective, and what are their limitations compared to other architectures?**Or****1B** Explain the backpropagation algorithm use in the neural network.**Q2.****2A.** Explain the Epoch and batch size. A model is trained for 5 epochs on a dataset with 50,000 samples using a batch size of 500. How many total iterations are required for training?**Or****2B.** Explain Overfitting and Underfitting with example.**Q3.****3A.** Discuss the different types of activation functions in AIML, and what are their specific uses?**OR****3B.** Explain the AIML project life cycle in details**Q4.****4A.** Discuss the main difference between Stochastic Gradient Descent, Batch Gradient Descent, and Mini-Batch Gradient Descent?**OR****4B.** Explain the concept of hyperparameter tuning in AIML with GridSearchCV?**Q5.****5A.** Explain the concept of Stemming in NLP with suitable example.**OR****5B.** Discuss Lemmatization. Why is Lemmatization considered more accurate?

Section B: All questions in this section are compulsory. Attempt either A or B part of each question. Each question carries 05 marks **[3QX5=15 Marks]**

Q6.

6A. Consider a simple convolution operation with the following parameters: Input Image: 5 by 5, Filter: 3 by 3, stride:1 and no padding. Given the Input Image represented by X and the filter represented by K. **[5 Marks]**

$$X = \begin{bmatrix} 1 & 2 & 3 & 0 & 1 \\ 4 & 5 & 6 & 1 & 2 \\ 7 & 8 & 9 & 2 & 3 \\ 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 0 \end{bmatrix} \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

- Compute the output feature Map size.
- Find the output feature map after performing the convolutional operations.

OR

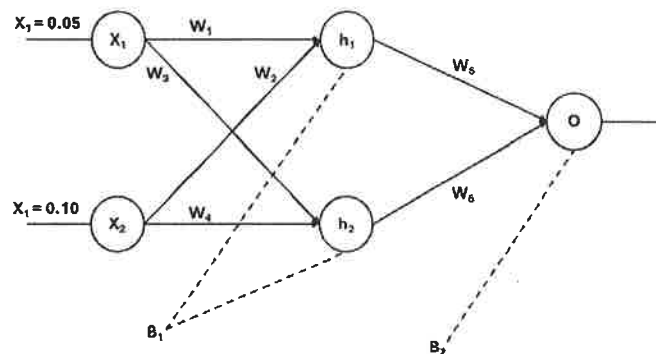
6B. Given the points A(3, 7), B(4, 6), C(5, 5) and D(6, 4), Analyse the core points, border point and outliers using DBSCAN. **[5 Marks]**

Take Eps = 2.5 and MinPts = 3 (Use any distance measures)

Data Points	X	Y
A	3	7
B	4	6
C	5	5
D	6	4

Q7.

7A. Consider a simple feedforward neural network with the following architecture: 2 nodes in the Input Layer, 1 hidden layer with 2 neurons with ReLU Activation Function, 1 output neuron with Sigmoid Activation Function, and Mean Squared Error (MSE) as loss function. The Target Output is 0.8. Weight and Bias information are: $w_1=0.5$, $w_2=-0.27$, $w_3=0.7$, $w_4=0.2$, $w_5=0.3$, $w_6=0.05$, $B_1=0.1$ and $B_2=0.05$. **[5 Marks]**



- Compute the Hidden Layer O/P.
- Calculate final output of the ANN.
- Find the Error.

OR

7B. Given the following 2D dataset with 4 data points:

[5 Marks]

X1	2	0	3	1
X2	0	2	1	3

Using PCA, compute the first principal component of the dataset.

Q8.

8A. Consider the following documents:

[5 Marks]

Review 1: I love cats

Review 2: I adore dogs

Review 3: I love both cats and dogs

Demonstrate the Bag of Words (BoW) representation for the given documents.

OR

8B. Consider the following documents:

[5 Marks]

Review 1: She enjoys reading books

Review 2: He watches movies daily

Review 3: They enjoy both books and movies

Calculate the term frequency (TF) and inverse document frequency (IDF) for the given documents and compute the TF-IDF representation for each document.

Section C: Compulsory question

[1Qx10 = 10 Marks]

Q9.

A CNN architecture is given. The input image is (32,32,1). Provide the output shapes for each layer in the architecture, assuming there is no padding. Also, write the total number of parameters.

Layer Type	Layer Details	Output Shape	Number of Parameters
Convolutional	Conv2D (8 filters, 5x5)		
Convolutional	Conv2D (16 filters, 3x3)		
Max Pooling	MaxPooling2D (2x2)		
Flatten	Flatten		
Dense	Dense (256 units)		
